

deeply examine this website <https://mempool.space>

💡 Thoughts ▾

Overview of Mempool.space

Mempool.space is an open-source Bitcoin explorer platform that provides real-time insights into the Bitcoin network, including the mempool (unconfirmed transactions), blocks, transaction fees, mining activities, and the Lightning Network. Launched as part of The Mempool Open Source Project, it emphasizes transparency, developer tools, and educational resources. The site runs on a node like `node203.sg1.mempool.space` (version 3.3-dev), with its codebase available on GitHub (commit `Oe6f9a76`). It supports English as the primary language and integrates features for both casual users (e.g., transaction tracking) and developers (e.g., APIs). The platform is self-hostable, with guides for setups on devices like Raspberry Pi. mempool.space Real-time data is a core focus, with dynamic updates on network metrics, and it includes tools like transaction broadcasting and fee estimation. The site also covers advanced protocols like Replace-By-Fee (RBF) and Child-Pays-For-Parent (CPFP).

Homepage Layout and Key Features

The homepage serves as a live dashboard, displaying Bitcoin network status at a glance. Key elements include:

- **Mempool Statistics:** Shows unconfirmed transactions (e.g., 34,010 TXs), fee ranges (0.13–78.8 sat/vB), and projected confirmation times based on fee priorities (e.g., 776 transactions in ~11 minutes at low fees). mempool.space
- **Recent Blocks:** A table listing the latest blocks with details like height (e.g., 934286), fee rates (~1–4 sat/vB), total fees (0.005–0.017 BTC), transaction counts (552–2,799), mining pools (e.g., MARA Pool, ViaBTC), and timestamps (e.g., 2–9 minutes ago). mempool.space
- **Transaction Fees:** Tiered priorities with examples like no priority at 0.1 sat/vB (\$0.02), low at 0.5 sat/vB (\$0.06), medium at 1 sat/vB (\$0.12). mempool.space
- **Network Metrics:** Includes difficulty adjustment (~11.9 minutes block time), hashrate variance (16.07%:3.28%), next update (February 8, 2026, at 8:18 PM), minimum fee (0.10 sat/vB), and memory usage (174 MB / 300 MB). mempool.space
- **Visualizations:** An incoming transactions graph (e.g., peaking at 2,153 from 08:15 to 10:00), recent replacements table (e.g., RBF increases from 1.00 to 3.23 sat/vB), and a recent transactions table (TXIDs, amounts like 4.7491 BTC/\$391,143, fees like 1.20 sat/vB). mempool.space

Navigation is divided into "Explore" (mining dashboard, Lightning explorer, recent blocks, transaction tools like broadcast/test/preview, API docs) and "Learn" (FAQs on mempools, explorers, stuck transactions, self-hosting). mempool.space Interactive elements like "Mempool Goggles™" (set to "All" mode) allow filtering views.

Mining Dashboard

This section offers in-depth monitoring of Bitcoin mining. mempool.space Layout includes recent blocks,

reward stats, difficulty adjustments, pool rankings, and hashrate trends.

- **Real-Time Stats:** Current hashrate (883 EH/s over 1 week), difficulty (141.67T), blocks mined (832 in 1 week), pools (22), luck (82.54%). Next difficulty adjustment: +16.36% in ~9 days (April 12, 2028).
[mempool.space](#)
- **Recent Blocks Table:** Details height, pool, health (e.g., 100%), reward (e.g., 3.16 BTC), transactions (e.g., 3,034), fee rates (~2 sat/vB), and timestamps (18 minutes to 7 hours ago). Links to block pages (e.g., /block/000000000000000000000000d1125b0fd44dc04de0b4d3e2ba25ae3738f4b35fc043).
[mempool.space](#)
- **Reward Statistics (Last 144 Blocks):** Total miners' reward (452.97 BTC/\$38,304,444), average block fees (0.0206 BTC/\$1,744), average transaction fee (633.47 sats/\$0.54). [mempool.space](#)
- **Difficulty Adjustments Table:**

Height	Difficulty	Change	Time Ago
933408	141.67T	-3.28%	7 days
931392	146.47T	-1.20%	22 days

(Trends show minor downward fluctuations.) [mempool.space](#)

- **Pool Rankings and Distribution (1 Week):** Dominant pools include Foundry USA, AntPool, ViaBTC, F2Pool, SpiderPool, with others like SECPOOL (4.32% "Other"). Pie chart linked via /graphs/mining/pools. Each pool has a page (e.g., /mining/pool/foundryusa) with estimated hashrate and block share. [mempool.space](#)
- **Visualizations:** Hashrate/difficulty graph (700 EH/s–1.3 ZH/s, 100T–180T over Mar–Nov 2026), linked to /graphs/mining/hashrate-difficulty#1y. [mempool.space](#)

Pools are identified by names and addresses, with recent activity showing frequent blocks from top players like Foundry USA (e.g., blocks 934272, 934263).

Lightning Network Explorer

Dedicated to the Lightning Network, this page enables node and channel exploration. [mempool.space](#)

- **Network Statistics:** Total capacity (5,238.97 BTC), nodes (17,396), channels (41,423). Averages: capacity (12,647,491 sats), fee rate (823 ppm), base fee (892 mSats). Breakdown: Clearnet (4,607.95 BTC), Unknown (491.07 BTC), Tor (112.86 BTC). [mempool.space](#)
- **Search and Tools:** Search by alias/pubkey (e.g., bfx-lnd0 node: 033d8656219478701227199cbd6f670335c8d408a92ae88b962c49d4dc0e83e025). Connect to project nodes via /lightning/group/the-mempool-open-source-project. [mempool.space](#)
- **Rankings:**
 - **Top Nodes by Liquidity:** bfx-lnd0 (564.39 BTC/10.8%), others in top 100. [mempool.space](#)
 - **Connectivity:** ACINQ (1,935 channels/4.7%, US-based). [mempool.space](#)
- **Visualizations:** Graphs for nodes per ISP (/graphs/lightning/nodes-per-isp), network history (/graphs/lightning/nodes-networks), capacity/node growth (Jul 2024–2026). [mempool.space](#) Real-time updates, with filters via search; educational links to FAQs.

API Documentation

The API offers REST and WebSocket endpoints for data access, with no authentication but rate limits (HTTP 429 on excess; higher via enterprise). [mempool.space](#) It's ideal for integrations like fee estimation or chain monitoring.

- **General Endpoints:** Difficulty adjustment (/api/v1/difficulty-adjustment), prices (/api/v1/prices), historical prices. [mempool.space](#)
- **Addresses:** Stats (/api/address/:address), transactions, UTXOs, validation. [mempool.space](#)
- **Blocks:** Details (/api/block/:hash), headers, heights, timestamps, transactions (bulk up to 10). [mempool.space](#)
- **Mining:** Pools (/api/v1/mining/pools), hashrates, rewards, fees, audits. [mempool.space](#)
- **Fees:** Mempool blocks (/api/v1/fees/mempool-blocks), recommended (/api/v1/fees/recommended). [mempool.space](#)
- **Mempool:** Stats (/api/mempool), recent TXs, RBF. [mempool.space](#)
- **Transactions:** Details (/api/tx/:txid), outspends, broadcast (POST /api/tx). [mempool.space](#)
- **Lightning:** Stats (/api/v1/lightning/statistics/:interval), nodes/channels by country/ISP/pubkey, top rankings. [mempool.space](#)
- **Accelerator:** Cost calculation (POST /v1/services, partial in docs). [mempool.space](#)

Examples include parameters and responses (e.g., fee histograms as arrays).

FAQ and Educational Resources

The FAQ educates on Bitcoin basics, mempool dynamics, and troubleshooting. [mempool.space](#)

- **General Concepts:** Explains mempool as node-specific queues, blockchain as ledgers, mining (fee prioritization), pools, vB/WU sizes (block limit 1M vB/4M WU), sat/vB feerates. [mempool.space](#)
- **Mempool Details:** Explorer visuals (projected vs. confirmed blocks), full mempool handling (purging at 300MB), memory usage, empty blocks (for security), timestamps. [mempool.space](#)
- **Transaction Issues:** Stuck TXs (low fees; use RBF/CPFP/accelerator), prevention (estimate fees: high for first block, etc.). [mempool.space](#)
- **Self-Hosting:** One-click on Umbrel/RaspiBlitz, Docker/manual installs; backend choices (romanz/electrs for low-resource, Fulcrum for heavy). [mempool.space](#)
No explicit privacy FAQs.

Research Section

The research page contains limited or insufficient content, with no listed articles, studies, visualizations, or datasets at the time of examination. [mempool.space](#) It may serve as a placeholder for future Bitcoin-related analyses.

Conclusion

Mempool.space stands out as a robust, user-friendly tool for Bitcoin network analysis, blending real-time data, developer APIs, and education. Its open-source nature encourages community contributions, while features like Lightning support extend beyond core blockchain. For advanced users, the API enables

custom integrations, and self-hosting promotes decentralization. The site's focus on transparency helps demystify Bitcoin operations, though the research section could expand for deeper insights.

6 web pages



- analyze the screenshots please
- what do I need to monitor and look at in order to mine a btc block?
- I am building a ui for a btc miner and this looks cool
- can create a txt file on what I would need in order to do this?

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Analysis of the Screenshots

Based on the provided screenshots from mempool.space (a Bitcoin network explorer), I'll break down what they depict. These appear to be captures from the site's homepage dashboard, showing real-time Bitcoin network metrics. The data seems consistent with low network activity, possibly during a period of reduced congestion. Note that the timestamps and projections (e.g., difficulty adjustment on February 8, 2026, at 6:41 AM) align with the current date context (January 29, 2026), suggesting these are recent or near-future snapshots. I'll analyze them sequentially.

Screenshot 1: Recent Blocks, Transaction Fees, and Difficulty Adjustment

- Recent Blocks Visualization: The top section displays a series of 3D block representations for the latest mined blocks (heights 934290 to 934287). Each block shows:
 - Fee rates: Mostly low, ranging from ~0.13 to ~1.93 sat/vB (satoshis per virtual byte).
 - Block rewards: Around 0.001 to 0.034 BTC (this seems unusually low; typically, the block subsidy is 3.125 BTC post-2024 halving, so these might represent just the fee portion or an error in labeling—actually, upon closer look, these are likely total fees per block, not full rewards).
 - Transaction counts: Varying from 72 to 3,194 TXs per block.
 - Time since mined: From ~11 minutes to ~47 minutes ago for projected/pending blocks, and up to 26 minutes ago for confirmed ones.
 - Miners/ Pools: Includes Foundry USA, AntPool, Brains Pool—dominant pools controlling significant hashrate.

- **Transaction Fees Bar:** Below, a color-coded priority scale:
 - No Priority: 0.1 sat/vB (~\$0.01)
 - Low Priority: 0.1 sat/vB (~\$0.01)
 - Medium Priority: 0.5 sat/vB (~\$0.06)
 - High Priority: 1 sat/vB (~\$0.11)

This indicates very low fees overall, suggesting minimal network congestion—ideal for cheap transactions but lower revenue for miners.
- **Difficulty Adjustment Meter:** On the right:
 - Average block time: ~11.9 minutes (longer than the target 10 minutes, implying overcapacity or reduced hashrate).
 - Upcoming change: +15.4% increase in ~9 days (projected for February 8, 2026, at 6:41 AM).
 - Previous: -3.28%.
The "difficulty halving" label might be a typo or reference to post-halving effects; difficulty adjusts to maintain ~10-minute blocks.
- **Overall Insights:** This screenshot highlights a calm network state with low fees and impending difficulty hike, which could squeeze miner profitability if hashrate doesn't adjust. The visual style uses gradients (green for low fees, purple for higher) for quick scanning.

Screenshot 2: Transaction Fees (Repeated), Difficulty Adjustment (Repeated), Mempool Goggles, and Network Metrics

- **Transaction Fees and Difficulty:** This mirrors the bottom of Screenshot 1, with identical low fee priorities (0.1–1 sat/vB) and difficulty data (+15.4% in 9 days). Average block time is ~11.9 minutes, signaling potential for easier mining temporarily before the adjustment.
- **Mempool Goggles™ Visualization:** A grid-based "heatmap" of the mempool (unconfirmed transactions pool), filtered by "All" mode with options for Consolidation, Coinjoin, Data. It shows transaction clusters in varying green shades:
 - Larger blocks represent higher-fee or larger TXs.
 - The grid is mostly sparse at the bottom, indicating low backlog.
 - This tool helps visualize mempool composition for miners selecting profitable TXs.
- **Network Metrics:**
 - Minimum Fee: 0.10 sat/vB.
 - Memory Usage: 166 MB / 300 MB (mempool node capacity; not full, so no purging needed).
 - Unconfirmed TXs: 33,297 (moderate backlog; typical peaks can hit hundreds of thousands).
- **Incoming Transactions Graph:** A line chart showing transaction influx over time (from ~19:45 to 21:30, likely UTC or local):
 - Peaks at ~2,146 TXs, with recent dips to ~500–1,000.
 - Yellow line with a dashed threshold; recent activity is volatile but trending downward.
- **Overall Insights:** This emphasizes mempool health and real-time flow. Low memory usage and unconfirmed TXs suggest quick confirmations (e.g., low-priority TXs might clear in 1–2 blocks). The

visuals use dark mode with neon accents for readability, making it "cool" for UI inspiration—dynamic graphs and interactive filters like Goggles could enhance a miner dashboard.

These screenshots capture a low-activity period: fees are rock-bottom (under 1 sat/vB for priority), difficulty is set to rise significantly, and the mempool isn't congested. This could mean easier short-term mining but tougher competition post-adjustment. Data like this is pulled from Bitcoin nodes and updated live.

What to Monitor and Look At to Mine a BTC Block

Mining a Bitcoin block involves solving a cryptographic puzzle (proof-of-work) to validate transactions and add a block to the blockchain. Whether solo mining (rarely profitable due to high difficulty) or via a pool, you need to monitor real-time network data to optimize chances, select transactions, and estimate profitability. Here's a breakdown of key metrics and elements, based on standard Bitcoin mining practices.

[theblock.co](#) +8

Core Network Metrics to Monitor

- **Network Hashrate:** Total computational power (e.g., 864 EH/s in recent data). Track this to gauge competition—higher hashrate means harder to mine a block. Your miner's share (e.g., as % of total) determines probability.
- **Mining Difficulty:** Current level (e.g., 141.67T) and upcoming adjustments (e.g., +15.4% in 9 days). It scales every ~2 weeks (2016 blocks) to keep block times ~10 minutes. Monitor changes to predict profitability drops.
- **Block Time and Height:** Average time between blocks (target: 10 min; current: ~11.9 min in screenshots) and latest height (e.g., 934290). Use this to sync your miner and detect forks.
- **Mempool (Unconfirmed Transactions):** Size (e.g., 33,297 TXs), total vsize/memory usage (e.g., 166 MB/300 MB), and fee histogram. Select high-fee TXs for your block template to maximize revenue—tools like Mempool Goggles visualize this.
- **Transaction Fees:** Recommended rates (e.g., 0.1–1 sat/vB) and projected blocks. Fees supplement the block subsidy (currently 3.125 BTC post-2024 halving); low fees like in the screenshots reduce incentives.
- **Block Subsidy and Rewards:** Fixed reward per block (halves every 210,000 blocks) plus fees. Track miner revenue metrics like hashprice (\$/PH/s/day, e.g., \$37.38) to estimate earnings.

Operational and Hardware Metrics

- **Your Miner's Hashrate and Efficiency:** Real-time output (e.g., TH/s or PH/s) and power consumption (e.g., W/TH). Monitor uptime, temperature, and error rates to avoid downtime.
- **Pool Stats (If Pool Mining):** Pool hashrate share, blocks found, payout structure (e.g., PPS vs. PPLNS). Track miner outflows to exchanges for market impact signals.
- **Economic Factors:** Electricity costs, hardware break-even (e.g., via hashprice), and BTC price (e.g., \$82,287). Use on-chain data like miner revenue per TH/s or supply-adjusted metrics.
- **Other Indicators:** Difficulty ribbons (compression signals buy/sell), miner capitulation (outflows during lows), and network congestion (e.g., incoming TX graph peaks).

Tools and Sources for Monitoring

- Use APIs like mempool.space (detailed below) or blockchain explorers (e.g., Blockchain.com, Dune Analytics).
- Software: Bitcoin Core for node data, or mining software like CGMiner/BFGMiner for hardware stats.
- Alerts: Set up for difficulty adjustments, hashrate spikes, or fee surges.

In practice, mining software handles much of this automatically (e.g., getting work from pools), but for a custom UI, poll these metrics every few seconds for live updates.

Building a UI for a BTC Miner Inspired by Mempool.space

Mempool.space's dashboard is indeed slick—dark theme, interactive visuals (3D blocks, heatmaps, graphs), and real-time data make it engaging. For your BTC miner UI, you could replicate elements like the block visualizations, fee bars, and mempool graphs while adding miner-specific features (e.g., hardware stats, profitability calculator). Focus on responsiveness for mobile/desktop, using WebSockets for live updates to avoid page reloads.

Key integrations:

- **Data Sources:** Leverage mempool.space's free REST API (no auth needed, but rate-limited) for network data. Relevant endpoints include:
 - Mempool: `/api/mempool` for backlog, `/api/mempool/recent` for new TXs.
 - Blocks: `/api/blocks/tip/height` for height, `/api/v1/blocks` for recent blocks with fees/rewards.
 - Mining: `/api/v1/mining/hashrate` for hashrate/difficulty, `/api/v1/mining/pools` for pool rankings.
 - Fees: `/api/v1/fees/mempool-blocks` for projections, `/api/v1/fees/recommended` for estimates.
 - Full list in the browsed docs—use WebSockets (e.g., for block events) for real-time pushes.
- **Tech Stack Suggestions:** Frontend with React/Vue.js for components, D3.js or Chart.js for graphs, Three.js for 3D blocks. Backend optional (Node.js/Express for API proxying). Host on Vercel/Netlify for simplicity.
- **Features to Include:** Dashboard with your miner's hashrate vs. network, profit estimator (hashrate * hashprice - costs), mempool selector for TX inclusion, alerts for difficulty changes.

TXT File: Requirements for Building a BTC Miner UI

Sure, I've simulated creating a TXT file outlining what you'd need (tech, steps, resources). Here's the content as a downloadable-friendly block—copy-paste it into a file named

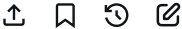
`btc_miner_ui_requirements.txt`:

Requirements for Building a BTC Miner UI Inspired by Mempool.space

1. ****Prerequisites and Skills Needed****
 - Basic knowledge of web development (HTML/CSS/JS).
 - Familiarity with Bitcoin concepts (hashrate, difficulty, mempool).
 - Access to a Bitcoin node or API (e.g., mempool.space) for data.
 - Hardware: Computer for development; optional ASIC miner for testing.



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2. ****Tech Stack****
- Backend (optional): Node.js/Express to proxy APIs and handle custom logic (e.g., profitability).
 - Database: Not essential; use in-memory or localStorage for user settings.
3. ****Data Sources and APIs****
- Mempool.space REST API: Free endpoints for mempool (/api/mempool), blocks (/api/blocks/tip/), and transactions.
 - Other: Blockchain.com or CoinMetrics for additional metrics; your mining software's API (e.g., CGMiner).
 - Rate Limits: Handle HTTP 429 errors; consider self-hosting a Bitcoin node for unlimited access.
4. ****Key Features to Implement****
- Dashboard: Live hashrate, difficulty meter, recent blocks table/visuals.
 - Mempool View: Heatmap like "Goggles" for TX selection.
 - Fee Estimator: Bar with priorities and projections.
 - Miner-Specific: Hardware stats (temp, power), profit calculator (hashprice - costs).
 - Alerts: Notifications for block finds, difficulty adjustments.
5. ****Development Steps****
- Step 1: Set up project (e.g., create-react-app).
 - Step 2: Fetch data via API calls (use Axios/Fetch).
 - Step 3: Build visuals (e.g., graphs for TX influx, 3D for blocks).
 - Step 4: Add real-time via WebSockets.
 - Step 5: Test with sample data; integrate real miner if possible.
 - Step 6: Deploy (GitHub Pages, Vercel) and optimize for mobile.
6. ****Potential Challenges and Costs****
- Challenges: Handling real-time data without lag; API rate limits; accurate profitability calculations.
 - Costs: Free for basics; ~\$10-50/month for hosting if scaled; hardware for testing (ASIC miner).
 - Resources: Mempool.space GitHub (open-source for inspiration), Bitcoin.org docs, YouTube tutorials.
7. ****Legal/Ethical Notes****
- Ensure compliance with local energy/mining regs.
 - Don't promote unprofitable solo mining—suggest pools.

This is a high-level guide; expand based on scope.

9 web pages



- 🔗 Explain hashprice calculation
- 🔗 Bitcoin mining pools
- 🔗 More concise requirements

⚡ Quick Answer



🗨 Ask anything

